

HUGO THOMAS

CIFRE Ph.D. student at Sorbonne Université, Ecole Normale Supérieure and Quandela

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EDUCATION

- 2023 – now **CIFRE (industrial grant) Ph.D. student in quantum machine learning for photonic**, Sorbonne Université, ENS and Quandela, Paris
Advisors: Elham Kashefi (LIP6), Ulysse Chabaud (DIENS), Pierre-Emmanuel Emeriau (Quandela)
- 2021 – 2023 **Master's degree in Quantum Information**, Sorbonne Université, Paris—*with highest honours*
Master's thesis: *Links between quantum circuits amplitudes and matrix permanents*
Master's thesis advisors: Rawad Mezher (Quandela) and Pierre-Emmanuel Emeriau (Quandela)
- 2018 – 2021 **Bachelor's degree in Computer Sciences**, Université de Paris-Diderot, Paris—*with highest honours*
First year at Université de Tours, Tours, France

RESEARCH

- Publications **Hugo Thomas**, Pierre-Emmanuel Emeriau, Elham Kashefi, Harold Ollivier and Ulysse Chabaud, *Role of coherence for quantum computational advantage*. Phys. Rev. Letters 135, 150602.
- Léo Monbroussou, Eliott Z Mamon, **Hugo Thomas**, Verena Yacoub, Ulysse Chabaud and Elham Kashefi, *Towards quantum advantage with photonic state injection*. Phys. Rev. Research 7, 033051.
- Preprints Benoit Seron, **Hugo Thomas**, Eduardo Araujo, Alex Arkhipov, Changhun Oh, Leonardo Novo, *Efficient classical algorithm for estimating linear statistics of Boson Sampling*.
- Léo Monbroussou, **Hugo Thomas**, Hela Mhiri, Zoë Holmes, Elham Kashefi, *Classical simulation and model concentration in passive linear optics*.
- Hela Mhiri, **Hugo Thomas**, Léo Monbroussou, Ulysse Chabaud, Zoë Holmes, Elham Kashefi, *Boson sampling beyond the dilute regime: second moments and anti-concentration*.
- Hugo Thomas**, Ulysse Chabaud and Pierre-Emmanuel Emeriau, *Shedding light on classical shadows: learning photonic quantum states*.
- Hugo Thomas**, Pierre-Emmanuel Emeriau and Rawad Mezher, *Connecting quantum circuit amplitudes and matrix permanents through polynomials*.

DISSEMINATION

Note: [‡] = presented by a coauthor.

- Talks [‡]*Boson sampling beyond the dilute regime: second moments and anti-concentration*, 9th International Conference on Quantum Techniques in Machine Learning (QTM26), Stellenbosch, South Africa
- [‡]*Boson sampling beyond the dilute regime: second moments and anti-concentration*, 26th Asian Quantum Information Science Conference (AQIS), Daejeon, Korea
- Shedding light on classical shadows: learning photonic quantum states*, Quantum Computing Theory in Practice (QCTiP26), Oxford, England.
- Shedding light on classical shadows: learning photonic quantum states*, 9th International Conference on Quantum Techniques in Machine Learning (QTM25), Singapore.
- Shedding light on classical shadows: learning photonic quantum states*, 8th International Conference for Young Quantum Information Scientists (YQIS25), Barcelona, Spain.

On the role of coherence for quantum computational advantage, the International Conference on Quantum Computing 2025 (ICoQC2025), Paris, France.

[‡]*Towards quantum advantage with photonic state injection*, the International Conference on Quantum Technology for High-Energy Physic (QT4HEP 2025), CERN, Switzerland.

On the role of coherence for quantum computational advantage, the 2nd Colloquium on Quantum Technologies (GdR-TeQ 2024), Paris, France.

Workshops [‡]*Towards quantum advantage with photonic state injection*, the 2nd Quantum Energy Initiative Workshop (QEI 2025), Grenoble, France (presenter: Léo Monbroussou).

Towards quantum advantage with photonic state injection, Quantum Software Lab Anniversary Workshop 2024, Edinburgh, U.K.

Posters *On the role of coherence for quantum computational advantage*, Quantum Computing Theory in Practice (QCTiP25), Berlin, Germany.

Connecting quantum circuit amplitudes and matrix permanents through polynomials, the 7th International Conference for Young Quantum Information Scientists (YQIS24), Paris, France.

Connecting quantum circuit amplitudes and matrix permanents through polynomials, Quantum Computing Theory in Practice (QCTiP 2024), the University of Edinburgh, U.K.

Quantum algorithms for matrix spectral approximation and interior point methods, 2022 Bad Honnef Summer School on Quantum Computing, Germany.

Invited seminar *On the role of coherence for quantum computational advantage*, LaBRI, Université de Bordeaux, France.

Public outreach Roundtable: “*Quantum revolution: the next tech breakthrough that no one understands (yet)*.”, Maison Mousse, Nantes, France

CEA Quantum Hub Discussion, CEA, France

STUDENTS

Feb. – July 2026 Thomas Delvaux (Master Student): Estimating nonlinear properties of photonic states via classical shadows.

REVIEW ACTIVITY

Journal Quantum Journal, Physical Review A

Conferences QTML 2024, QPL 2025, QIP 2025, TQC 2026, QPL 2026

INTERNSHIPS

June – Aug. 2022 Quantum algorithms for matrix approximation and interior points methods, IRIF, Paris
Advisor: Simon Apers

SUMMER SCHOOLS

June 2025 [LMS Research School Quantum Machine Learning and Hamiltonian Simulation](#), Sabhal Mor Ostaig, Isle of Skye

August 2022 [Bad Honnef Summer School on Quantum Computing](#), Physikzentrum Bad Honnef, Germany